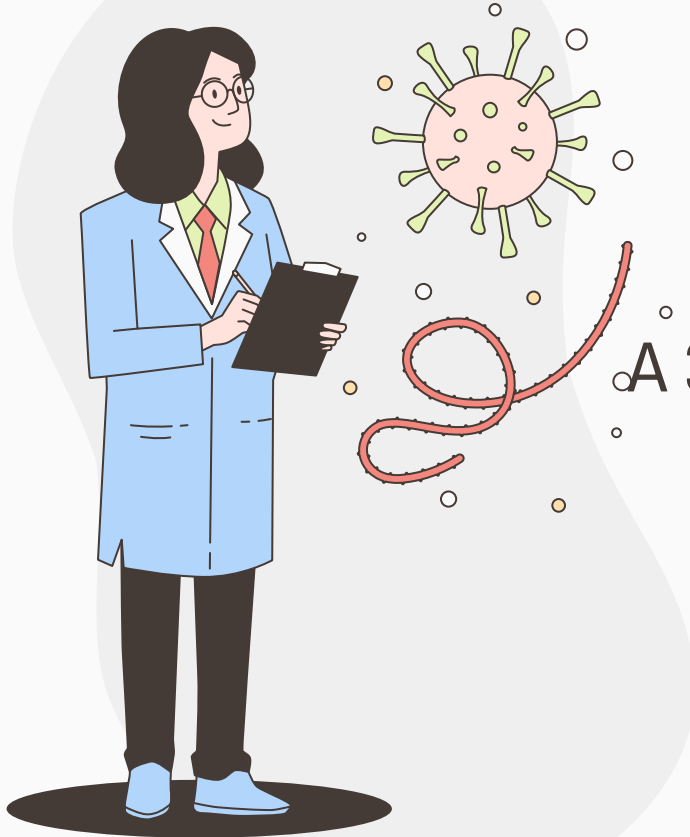




Interesting Case

A 3 year-old boy with subacute fever
with rash for 1 week

Pattranan Kusontammarat, MD

**Division of Pediatric Infectious Diseases,
Department of Pediatrics
Faculty of Medicine, Chiang Mai University**

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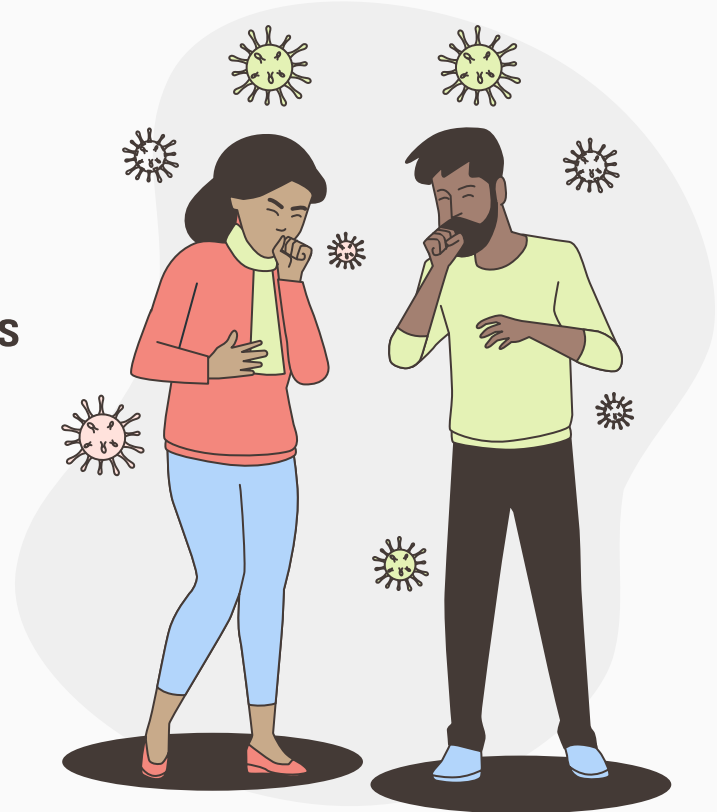
Case Presentation

02

Reactive Infectious Mucocutaneous
Eruption (RIME)
Erythema Multiforme (EM)

03

EBV Infection & EBV induced HLH



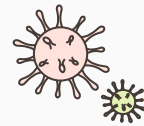


Chief Complaint

- ผู้ป่วยเด็กชาย ชาวไทย อายุ 3 ปี 7 เดือน
- ภูมิลำเนา อ.สารภี จ.เชียงใหม่
- Admission date 2/1/68

Identification Data

Subacute fever with rash for 1 weeks

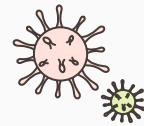


Present Illness



19/12/67

เริ่มมีไข้ต่ำๆ มีอาการไอ
น้ำมูก ซื้อยาร้านขายยามา
ทาน เป็นยาลดไข้ แก้แพ้ วัน
ต่อมาเริ่มมีผื่นแดงขึ้นที่แก้ม



Present Illness



19/12/67

เริ่มมีไข้ต่ำๆ มีอาการไอ
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23/12/67

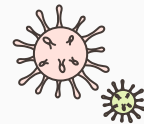
ยังมีไข้ มีอาการไอ น้ำมูกมากขึ้น มีผื่นนูนแดงขึ้น
ที่แก้ม มีปากแดง ไม่มีแผลในปากไม่มีผื่นที่ตัว
visit รพ สารภี PE: **BT 39.2** CBC Hb 11 g/dl
Hct 33.4 % **WBC 4,310** cell/mm³ N 63%
L 28% platelet 194,000 cell/mm³
Tx: oral amoxycillin + CPM

19/12/67



23/12/67





Present Illness

Admit 25/12/67-2/1/68



19/12/67

เริ่มมีไข้ต่ำๆ มีอาการไอ
น้ำมูก ซ้ำยาร้านขายยามา
ทาน เป็นยาลดไข้ แก่แพ้ วัน
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23/12/67

ยังมีไข้ มีอาการไอ น้ำมูกมากขึ้น มีผื่นนูนแดงขึ้น
ที่แก้ม มีปากแดง ไม่มีแผลในปากไม่มีผื่นที่ตัว
visit รพ สารภี PE: **BT 39.2** CBC Hb 11 g/dl
Hct 33.4 % WBC 4310 cell/mm³ N 63% L
28% platelet 194,000 cell/mm³
Tx: oral amoxycillin + CPM



25/12/67

Revisit รพ สารภี เนื่องจากยังมีไข้ มีผื่นมากขึ้นที่ปาก ที่
แขนขา ไม่มีผื่นที่ตัว ผื่นคุ่มวงมากขึ้น ไม่มีมือเท้าบวม ไม่มี
ตาแดง ยังมีไอมากอยู่ ไม่มีหายใจเหนื่อย
PE: **maked injected pharynx**, no exudate,
purplish rash at mouth, no injected conjunctiva
Lungs: clear and equal BS
Heart: no murmur
Skin: **purplish rash both cheek, arms and legs**



Present Illness

Admit รพ สารภี 25/12/67 W/U Lab

- **CBC:** Hb **10.9** g/dl Hct 32.9 % WBC **3,610** cell/mm³ (N 51% L 41% Eo 2.8%) plt 228,000 cell/mm³
- **LFT:** TP 5.9 g/dl Alb 3.8 g/dl Glob 2.03 g/dl **AST 220 U/L ALT 57 U/L** TB 0.39 mg/ml DB 0.08 mg/ml
- **Electrolyte:** BUN 6.5 mg/dl Cr 0.29 mg/dl Na 135 mmol/l K 3.9 mmol/L Cl 103 mmol/l CO2 36 mmol/l

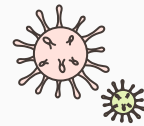
Impression: subacute fever with purplish rash with transaminitis DDx SJS, SSSS, Scrub typhus

Management :

- 5% DN/2 IV rate 50 ml/hr (MT)
- Empirical antibiotic Cefotaxime (150 MKDay) 700 mg IV q 8 hr (25-29/12/67)
- Consult CMU 26/12/67







Present Illness

Admit สารภี 25/12/67-2/1/68



26/12/67

Consult CMU

**Impression: Reactive infectious
mucocutaneous eruption (RIME)**

Suggest

- Treat mycoplasma infection
- Prednisolone 0.5 MKDay



27/12/67

ไข้เริ่มลง อาการดีขึ้น



29/12/67

หลังจากนั้นไข้มีขึ้นมาใหม่ และผื่นลามมากขึ้น
ตรงใบหน้า แขนขา ผ่ามือผ่าเท้า ไม่มีผื่นใน
ปาก ไม่มีตาแดง ไม่มีผื่นตามลำตัวและหลัง
อาการอื่นปกติ ทานได้น้อย เนื่องจากเจ็บปาก



2/1/68

Refer CMU

Cefotaxime (26-29/12/67)

Ceftazidime (29/12/67-2/1/68)

Azithromycin (26-30/12/67)



Past History

- Prenatal: ฝากครรภ์ที่รพ.แม่และเด็ก, ไม่มีภาวะแทรกซ้อนระหว่างฝากครรภ์
- Natal: : Term, BBW 2,195 gm, มีตัวเหลืองได้สองไฟแรกเกิด ได้นอนรพ. 7 วัน
- No Known U/D, no current medication, no food / drug allergy
- Vaccination: As EPI Programmed + vaccine เสริม influenza, PCV, HAV, VZV vaccine
- Growth and development: สมวัย
- Family history: ไม่มีประวัติโรคทางพันธุกรรมในครอบครัว

Physical Examination

At CMU:

Vital Signs	BT 37.2°C, HR 148 /min, RR 28, BP 107/80 mmHg, SpO ₂ 100%
Growth	Weight 13 kg g (P ₁₀₋₂₅) Height 95 cm (P ₁₀₋₂₅) BMI 14.4 kg/m ²
GA	Normal conscious
HEENT	No injected conjunctiva, mild injected pharynx , no oral or buccal ulcer multiple ulcers on upper and lower lips
Heart	Regular rhythm, no murmur, normal S1S2
Lung	No retraction, clear and equal BSBL

Physical Examination

At CMU:

Abdomen	No distension, soft, no HSM, no palpable mass
Extremities	Pulse full, CRT<2 secs
Neuro	good conscious
Skin	Generalized purplish target like lesions distributed on cheeks, lips, upper and lower extremities, involve palm & sole No involve oral or buccal mucosa, no involved perineum or perianus





Laboratory Investigation

Hb	9.7	g/dL
Hct	30.2	%
WBC	4750	cells/cu.mm.
Neutrophil	48.3	%
Eosinophil	0.2	%
Basophil	0.6	%
Lymphocyte	40.8	%
Monocyte	10.1	%
Platelet	317,000	cells/cu.mm
MCV	66.5	FL.

Serum iron	26	ug/dL
TIBC	377	ug/dL
TSAT	6.9	%

Laboratory Investigation

BUN	7	mg/dL
Creatinine	0.31	mg/dL
Na	135	mmol/L
K	4.5	mmol/L
Cl	101	mmol/L
Total CO2	19	mmol/L
Ca	9.1	mg/dl
P	4.3	mg/dl
Mg	1.74	mEq/L

Total protein	6.6	g/dL
Albumin	3.7	g/dL
Globulin	2.9	g/dL
ALP	212	U/L
Cholesterol	188	mg/dL
AST	317	U/L
ALT	75	U/L
TB	0.32	mg/dl
DB	0.11	mg/dl

Laboratory Investigation

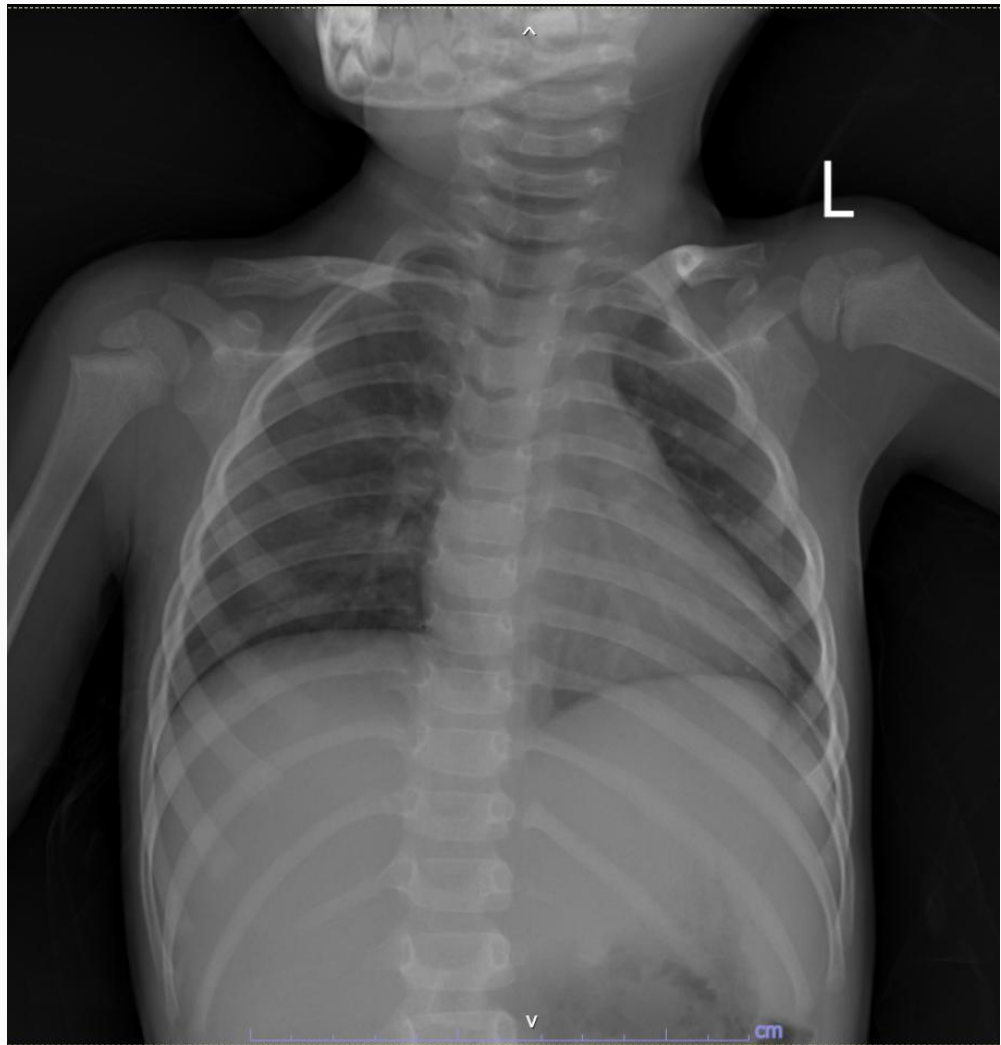
Mycoplasma IgM	Positive
Mycoplasma IgG	Positive
Anti-CMV IgM (EIFA)	Negative
Anti-CMV IgG (EIFA)	Positive 10
Herpes Simplex (I+II) IgM (ELISA)	Negative
Herpes Simplex (I+II) IgG (ELISA)	Positive

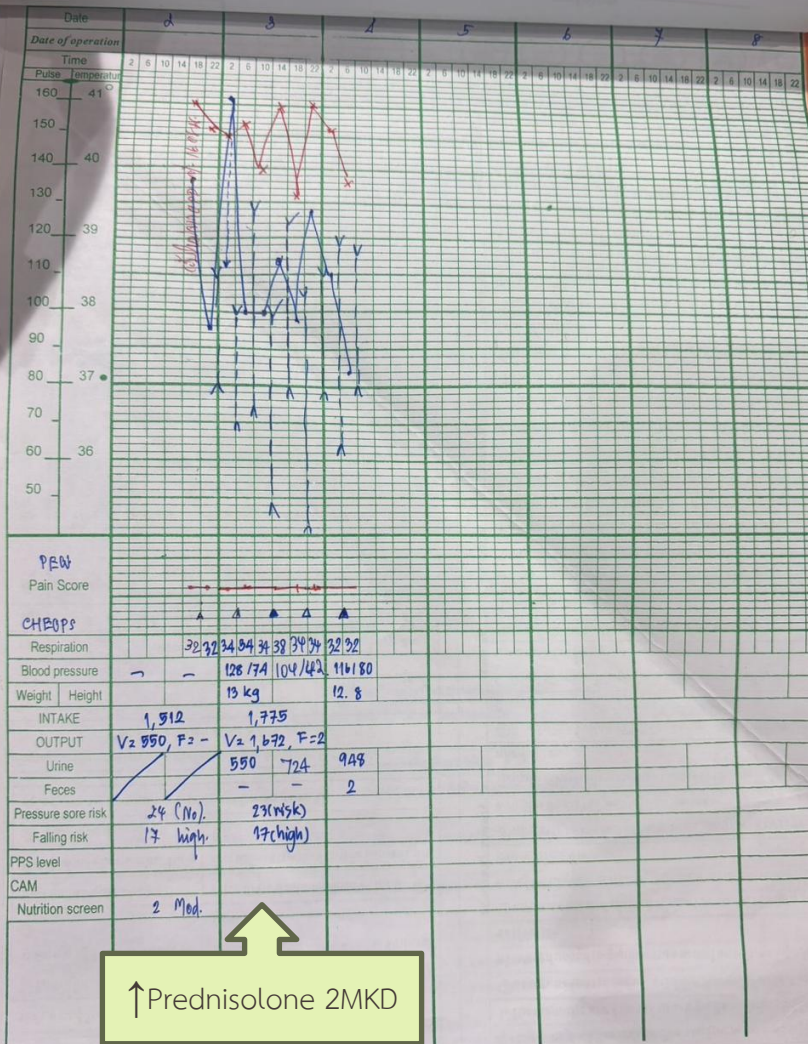
Lesion for Tzanck smear; not seen multinucleated giant cell

Laboratory Investigation

EBV VCA - IgM	Positive
EBV VCA - IgG	Positive
EBNA	Positive

CXR





Persistent fever- work up (3/1/68)

- ESR 26 mm/hr
- CRP 8.5
- Procalcitonin 0.533 ng/ml

Impression: Bullous Erythema Multiforme induced from Mycoplasma Infection

Management:

- Increase prednisolone 2 MKDay
- Dressing lesion with vasaline + mupirocin

Lab Investigation

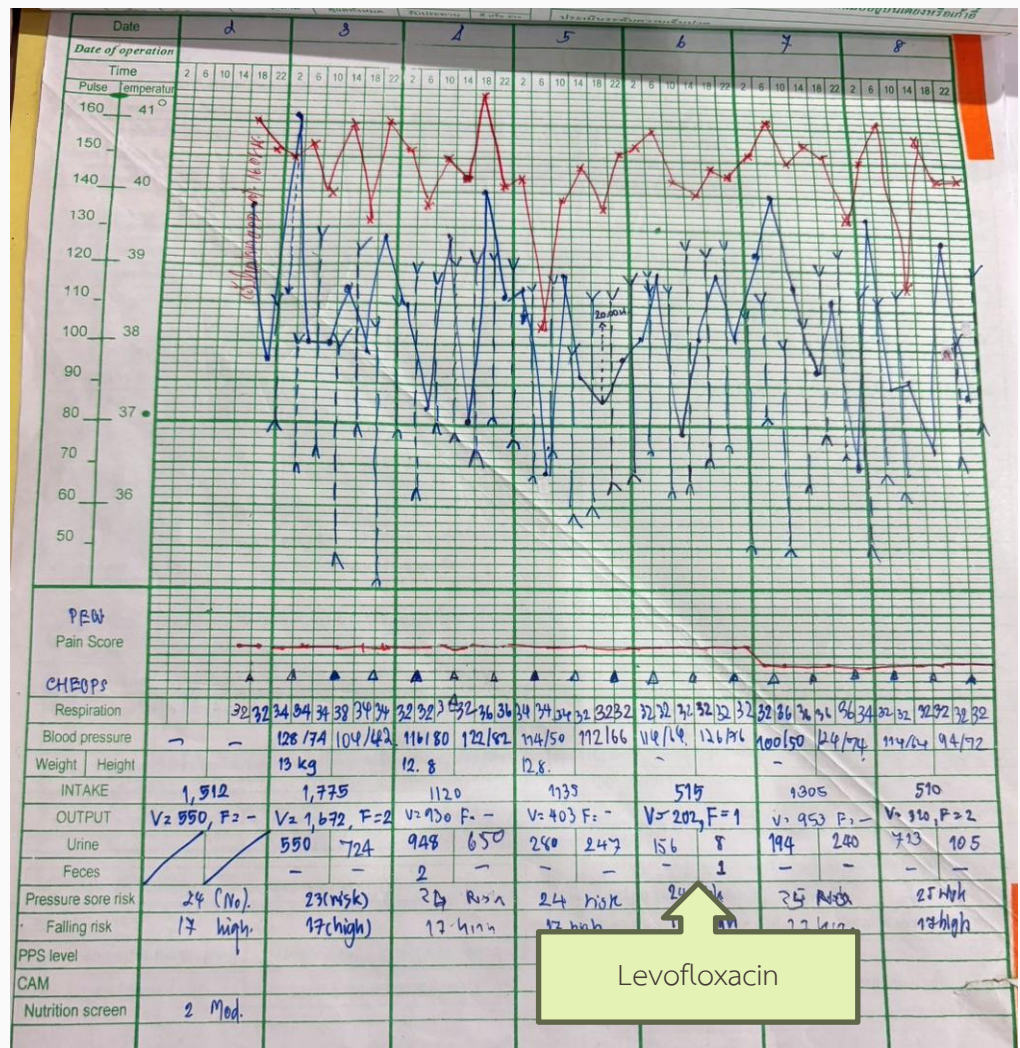


Mx: Consult hemato work up
bicytopenia (6/1/68)

CBC	6/1/68★	7/1/68	8/1/67
Hb (g/dl)	8.8	8.6	8.7
Hct (%)	26.1	26.3	26.1
WBC (cell/cu.mm.)	2210	3,420	3,040
Neutrophil (%)	54	58	66
Lymphocyte(%)	31	28	28
Atypical lymphocyte (%)	6	3	0
Monocyte(%)	8	8	5.3
Eosinophil (%)	0	0	0
ANC	1,193	1,983	2,006
Platelet(cell/cu.mm.)	328,000	349,000	356,000
MCV	64	64.5	64

Progression

- Suggest treat as drug resistance Mycoplasma infection
- Start levofloxacin IV (6/1/68)



Lab Investigation

RC	1.99	%
Ret-He	18.1	pg
IPF	2.8	%
cRC	1.40	%

Coomb test	
DAT	neg
IAT	neg

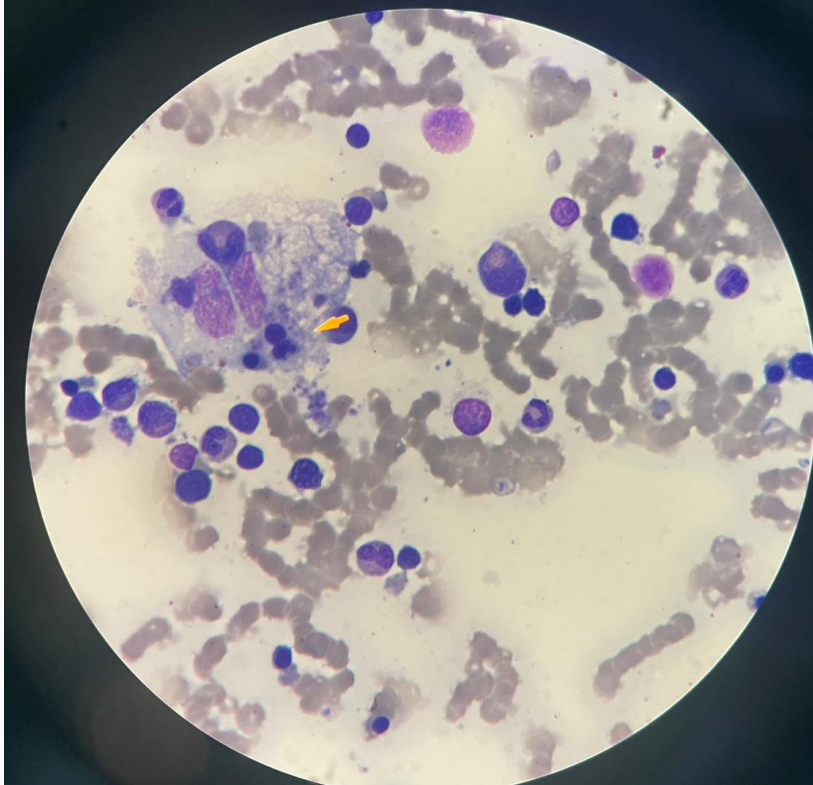
Lab Investigation

	2/1/68	6/1/68
Total protein(g/dl)	6.6	3.0
Albumin(g/dl)	3.7	3.5
Globulin(g/dl)	2.9	2.6
ALP (U/L)	212	192
Cholesterol (mg/dl)	188	192
AST (U/L)	317	326
ALT (U/L)	75	70
TB (mg/dl)	0.32	0.30
DB (mg/dl)	0.11	0.10

fibrinogen	113.9	mg/dL
ferritin	20,253	mg/dL
Triglyceride	228	mg/dL

PT	10.9	sec
PTT	32.7	sec
INR	1.03	

Lab Investigation



BMA: Seen hemophagocytic activity

Diagnosis Criteria: HLH-2004

A. Molecular diagnosis

Pathogenic biallelic mutation in genes associated with HLH: PRF1, UNC13D, STX11, STXBP2, LYST, RAB27A, XIAP, SH2D1A, and NLCR4

B. Clinical and laboratory criteria

For clinical diagnoses, greater than or equal to five of the eight criteria below must be met

1. Fever

2. Splenomegaly

3. Cytopenia

≥ 2 cell lines

- Hemoglobin

< 9.0 g/dL (neonate < 10 g/dL)

- Platelets

< 100,000 cells/cu.mm.

- Neutrophils

< 1,000 cells/cu.mm.

4. Hypertriglyceridemia

≥ 265 mg/dL (≥ 3 mmol/L)

Hypofibrinogenemia

≤ 150 mg/dL (≤ 1.5 g/L)

5. Hemophagocytosis

Bone marrow, other tissues

6. Hyperferritinemia

≥ 500 ng/mL (≥ 500 μg/L)

7. Reduced/absent NK cell cytotoxicity

8. Elevated soluble CD25

≥ 2,400 U/mL

R/O EBV Reactivation

EBV viral load (IU/ml)	1,000,000
EBV IgM (ELISA)	Positive
EBV IgG (ELISA)	Positive

Impression: EBV-induced HLH

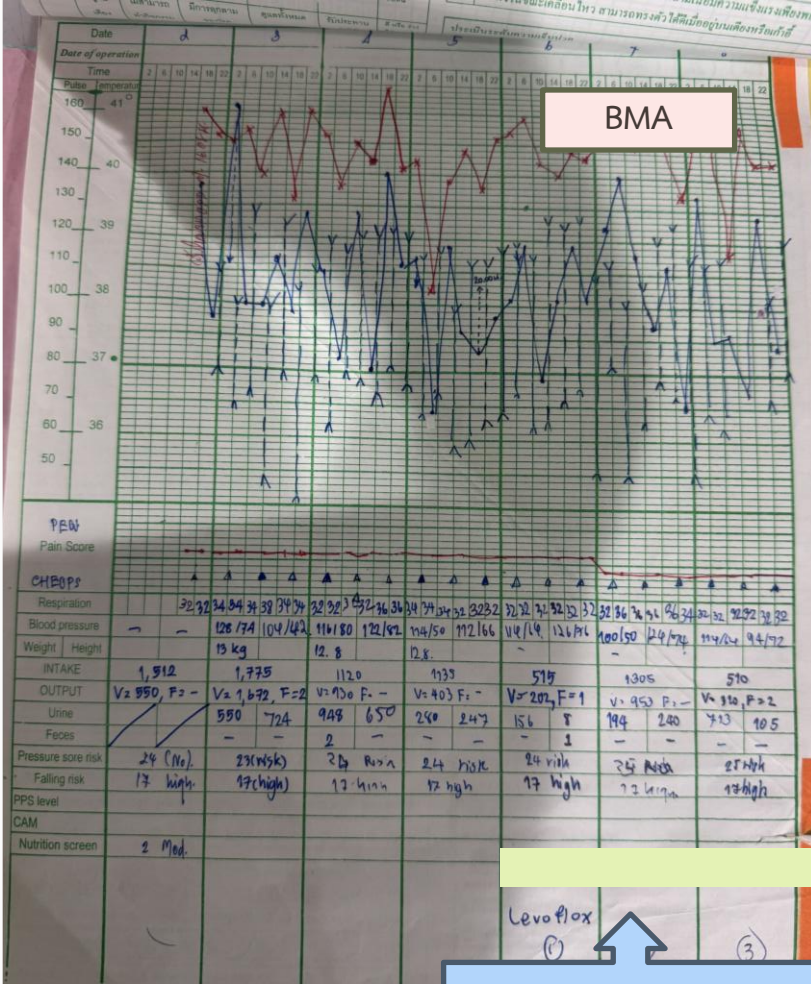
Management

- IVIG (1gm/kg) (total time 12 hr) 15 gm IV drip as protocol
- Dexamethasone (10mg/m²/day) 3 mg iv q 12 hr

Diagnosis

- Bullous EM trigger by *Mycoplasma* infection
- EBV induced HLH

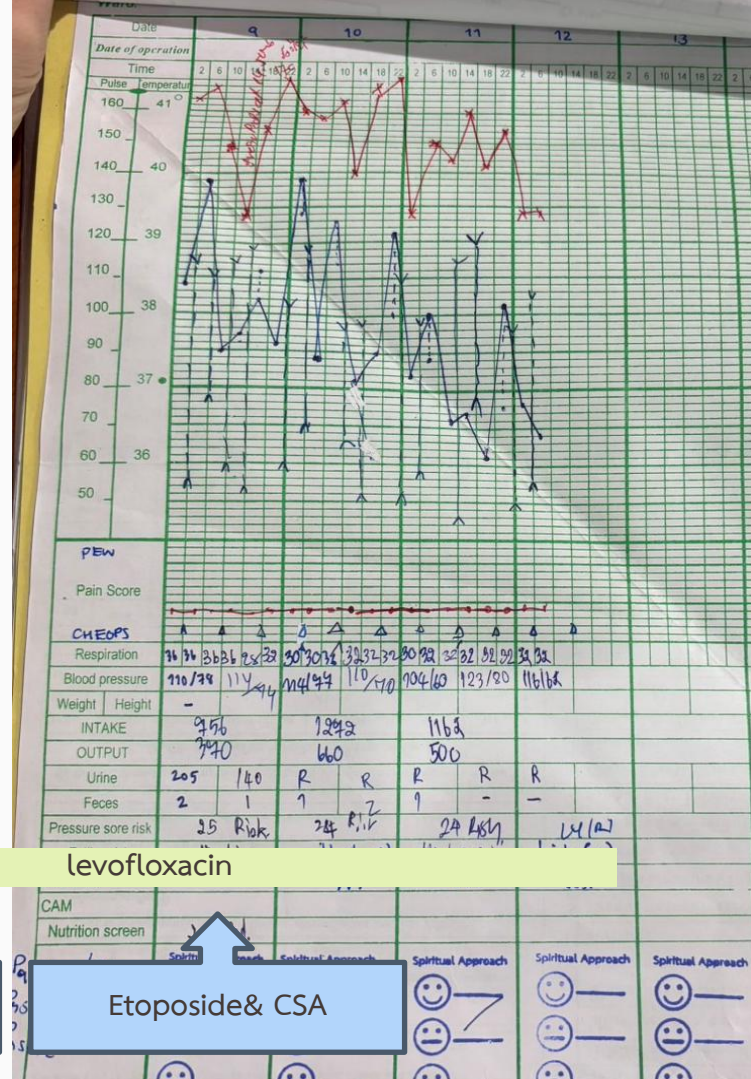




BMA

Levoflox
(P)

IVIG & Dexamethazone



levofloxacin

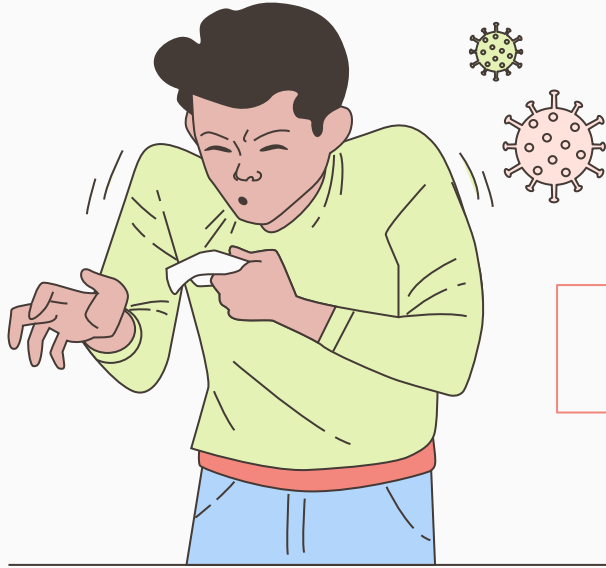
Etoposide & CSA

Skin Biopsy

Skin, right forearm, punch biopsy

- Basal vacuolization with death keratinocyte
- Necrotic epidermis separation from dermis, focal
- Superficial perivascular histiolytic infiltration

Clinical diagnosis : Bullous erythema multiforme Ddx SJS, CBDC, Rowell



Mycoplasma Infection

Extrapulmonary Manifestations

Mycoplasma Infection: Extrapulmonary Manifestation



Skin manifestation

- 10% of children with *M pneumoniae* infection will exhibit rash , most often maculopapular



Atypical manifestation

- **CNS:** aseptic meningitis, encephalitis, acute disseminated encephalomyelitis, cerebellar ataxia, transverse myelitis, peripheral neuropathy)
- **CVS:** myocarditis, pericarditis, arthritis (particularly in immunocompromised host)

Dermatological Manifestation

Common (> 1% of cases)

Exanthematous skin eruptions

Erythema nodosum

Urticaria

SJS (erythema multiforme major)^a

Rare

Bullous erythema multiforme (minor)^a

Pityriasis rosea

MP-associated mucositis^a

Henoch–Schönlein purpura

Toxic epidermal necrolysis^a

Leucocytoclastic vasculitis

Kawasaki disease

Subcorneal pustular dermatosis

Thrombotic thrombocytopenic purpura

Sweet syndrome

Raynaud phenomenon

Reiter syndrome

Urticarial vasculitis

Gianotti–Crosti syndrome



Reactive Infectious Mucocutaneous Eruption (RIME)

Formerly referred as *Mycoplasma* induced rash and
mucositis syndrome (MIRM)

Reactive Infectious Mucocutaneous Eruption (RIME)

- Severe mucocutaneous reaction most frequently seen in children and adolescents after an infectious respiratory illness, particularly *Mycoplasma pneumoniae* infections
- Characterized by prominent oral mucositis, conjunctivitis, and urethral involvement
- Cutaneous manifestations are usually limited; the most frequent are vesicobullous eruptions
- RIME is distinguished from Stevens–Johnson syndrome and toxic epidermal necrolysis because it typically has an infectious cause, a milder course, and a favorable prognosis

Reactive Infectious Mucocutaneous Eruption

Infection	Comment	References
Probable triggers		
<i>Chlamydomydia pneumoniae</i>	–	Mayor-Ibarguren <i>et al.</i> , 2017; ¹³ Duarte <i>et al.</i> , 2010 ³⁵
Influenza B	–	Goyal & Hook, 2010; ³⁶ Mazori <i>et al.</i> , 2020 ³⁷
Enterovirus/rhinovirus	Recurrent cases	Olson <i>et al.</i> , 2017 ³⁸
Human metapneumovirus		
Human parainfluenzavirus 2		
Possible triggers		
Varicella zoster virus	Review of all reports in reference 38	Bay <i>et al.</i> , 2005 ³⁹
Hepatitis A	Cholestatic hepatitis, possible effect on drug metabolism	Werblowsky-Constantini <i>et al.</i> , 1989 ⁴⁰
Epstein–Barr virus	–	Maggio <i>et al.</i> , 2011; ⁴¹ Brunet-Possenti <i>et al.</i> , 2013 ⁴²
Merkel cell polyoma virus	Immunosuppressed transplant patient with confirmed HSV1 mucositis	Maximova <i>et al.</i> , 2013 ⁴³
Cytomegalovirus	Patient was on high-dose prednisolone	Cruz <i>et al.</i> , 2010 ⁴⁴
HHV-6	Reactivation of HHV6 in the setting of drug-induced SJS to TMP/SMX or doxycycline	Neuman <i>et al.</i> , 2013 ⁴⁵

HHV, human herpesvirus; HSV, herpes simplex virus; SJS, Stevens–Johnson syndrome; TMP/SMX, trimethoprim/sulfamethoxazole.

Erythema Multiforme

- Self-limited hypersensitivity syndrome
- Occur at any age, most common in young adults; 20% of cases occur in children
- Majority of cases of EM in children: precipitated by herpes simplex virus (HSV) type 1 infection, although HSV type 2 infection has been described in affected adolescents
- Mycoplasma has also been associated
- Approximately 50% of cases follow herpes labialis infection, usually by 3 to 14 days, although concurrent infection and EM have been noted
- Lesions may also become bullous (**bullous EM**) but in contrast to epidermal necrolytic disorders, less than 2% of the body surface area is generally involved

Erythema Multiforme



Fig. 20.35 Erythema multiforme. Classic target lesions and margined wheals with central vesicles are characteristic.



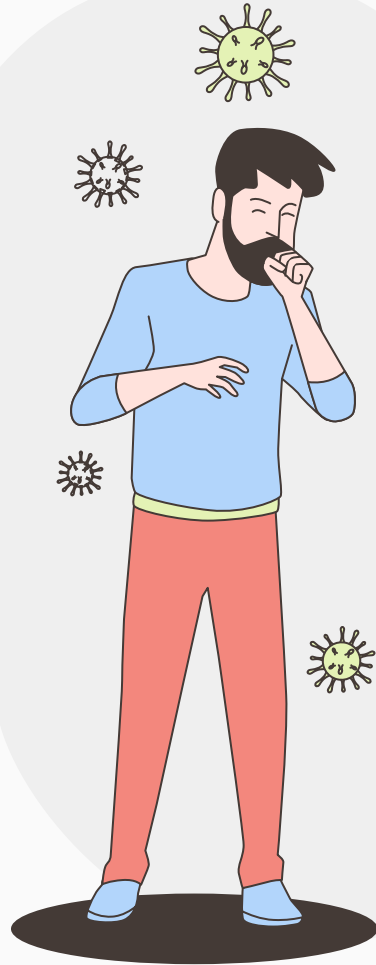
Fig. 20.36 Erythema multiforme. Extensive target lesions on the thighs of a 12-year-old girl. The recurrent reaction was suppressed by administration of oral acyclovir.

Hallmark: erythematous ring (iris or target lesion)

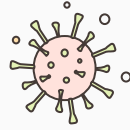
Table 20.3 Classification of Epidermal Necrolysis and Comparison with Erythema Multiforme and RIME

	Erythema Multiforme	RIME	SJS	SJS-TEN	TEN
Lesional morphology	Fixed typical or atypical target lesions	Vesiculobullous; no typical papular targets	Targetoid lesions, dusky red macules, bullae	Targetoid lesions, dusky red macules, bullae	Targetoid lesions, dusky erythematous macules and plaques; detachment of epidermis
Localization of skin lesions	Acral distribution is characteristic; can be widespread	May be no skin involvement or limited	May be scattered and isolated; may be confluent, especially on the trunk and face	May be scattered and isolated; often confluent	Usually extensive involvement with widespread confluence
Involved skin	<2%	<10%	<10%	10% to 30%	>30%
Biopsy features	Individual apoptotic keratinocytes, but not large sheets of epidermal necrosis	Interface dermatitis with dyskeratosis	More interface dermatitis	Significant interface dermatitis + necrosis	Predominantly necrosis
Mucosal changes	Tends to be milder	Predominant erosive mucositis	Prominent	Prominent	May be less than in SJS
Systemic involvement	Minimal or absent	Often present (cough, fever, malaise, arthralgias)	Often present	Always present	Always present
Most common underlying cause	Typically HSV	Most often <i>Mycoplasma</i> ; <i>Chlamydia</i> has been associated	Often drug-induced. Sometimes <i>Mycoplasma</i>	Usually drug	Most often drug; can be a feature of graft-vs.-host disease

HSV, Herpes simplex infection; *RIME*, reactive infectious mucocutaneous or mucosal-predominant eruption (includes *Mycoplasma* and *Chlamydia* induced); *SJS*, Stevens–Johnson syndrome; *TEN*, toxic epidermal necrolysis.



EBV Infection



EBV Infection

- **Infectious mononucleosis** is the most common presentation of primary symptomatic Epstein-Barr virus (EBV) infection
 - Fever, pharyngitis with or without petechiae and often with exudate, lymphadenopathy and hepatosplenomegaly
 - Atypical lymphocytosis and elevated hepatic aminotransferase concentrations are frequently found on laboratory evaluation
- Infections are typically unrecognized or nonspecific in infants and young children.
- **Rash can occur in up to 20% of patients** and is more common in patients treated with antibiotics, most commonly ampicillin or amoxicillin as well as with other penicillins



Skin Eruption of Infectious Mononucleosis



Fig. 16.38 Infectious mononucleosis. A diffuse, erythematous macular and papular eruption occurred in this adolescent male after receiving amoxicillin therapy.

- Nonspecific, may present as a macular, petechial, scarlatiniform, urticarial, or erythema multiforme–like rash
- After administration of ampicillin or amoxicillin, 90-95% of patients develop an erythematous macular and papular eruption that may be related to ampicillin-antibody immune complexes resulting from B-cell activation
- This reaction occurs about 5 to 9 days after starting the antibiotic and does not appear to represent a true drug allergy



Diagnosis

- **Infectious mononucleosis is primarily a clinical diagnosis**
- Serologic testing for EBV is useful, particularly for evaluating patients younger than 4 years or in whom the infectious mononucleosis syndrome is not classic
- IgM anti-VCA antibody and the absence or very low titers of antibodies to Epstein-Barr nuclear antigen (EBNA) is useful for identifying **active and recent infections**
- Antibody against EBNA is not present until several weeks to months after onset of infection and rises with convalescence, a very elevated anti-EBNA antibody concentration typically excludes active primary infection

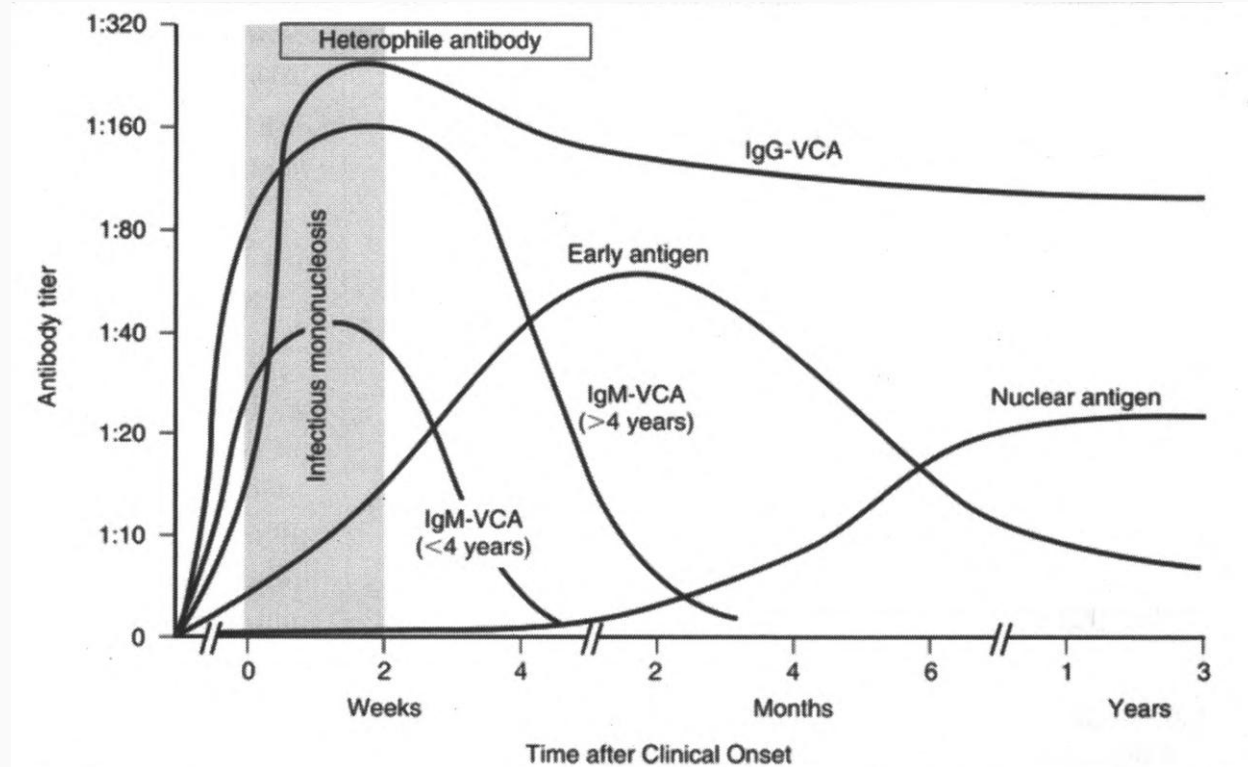


Laboratory Diagnosis

Infection	VCA IgG	VCA IgM	EA (D)	EBNA
No previous infection	—	—	—	—
Acute infection	+	+	+/-	—
Recent infection	+	+/-	+/-	+/-
Past infection	+	—	+/-	+

- Nucleic acid amplification tests for detection of EBV DNA in serum, plasma, whole blood, and tissue can be useful in evaluation of immunocompromised patients and in complex clinical situations
- EBV viral load seem particularly useful in diagnosing acute IM when the serologic studies are inconclusive

Evolution of Antibodies to Various Anti EBV Antigens



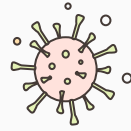
Hemophagocytic Lymphohistiocytosis (HLH)

Categorization of HLH: Primary and Secondary HLH.

- **Primary HLH:** Caused by genetic mutations
- **Secondary HLH:**
 - IAHS (Infection-Associated Hemophagocytic Syndrome)
 - MAHS (Malignancy-Associated Hemophagocytic Syndrome)
 - MAS (Macrophage Activation Syndrome), associated with autoimmune -
autoinflammatory diseases.

Treatment

- There is no antiviral treatment approved for EBV infection
- Patients suspected of having infectious mononucleosis should not receive ampicillin or amoxicillin, which may cause nonallergic morbilliform rashes
- Short-course corticosteroids: prednisone 1 mg/kg per day orally (maximum 60 mg/day), for 5 to 7 days, with tapering over 1 to 2 weeks considered if
 - Marked tonsillar inflammation with impending airway obstruction
 - Massive splenomegaly
 - Myocarditis
 - Hemolytic anemia
 - HLH or severe thrombocytopenia



EBV induced HLH

- Life-threatening HLH has been treated with cytotoxic agents and immunomodulators, including etoposide, corticosteroids, and/or cyclosporine
- **Emapalumab** is approved for the treatment of progressive or refractory HLH

Thank You

